Vidyavardhini's College of Engineering &Technology
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Experiment No.2
Aim: Implementation and Analysis of TCP/IP Utility Commands
Date of Performance: 22/07/2026
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Experiment No 2

AIM: Implementation and Analysis of TCP/IP Utility Commands

THEORY AND OUTPUT:

ipconfig for windows

Output

```
C:\Users\SHAHO>ipconfig

Windows IP Configuration


Ethernet adapter Ethernet:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 9:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 10:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   IPv6 Address. . . . . . . . . . . : 2402:3a80:1393:2d5b:ecdb:5bc7:7e58:839c
   Temporary IPv6 Address. . . . . . : 2402:3a80:1393:2d5b:e8e7:816c:d5de:cd1d
   Link-local IPv6 Address . . . . . : fe80::4c1a:87fb:4fa:2545%10
   IPv4 Address. . . . . . . . . . . : 10.109.253.201
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : fe80::c888:96ff:fe11:5d09%10
                                       10.109.253.62
```

ifconfig for Linux:

```
shahid@SHAHO:~$ ifconfig
enp4s0: flags=4099<UP,BROADCAST,MULTICAST>  mtu 1500
        ether 58:11:22:3f:41:ad  txqueuelen 1000  (Ethernet)
        RX packets 0  bytes 0 (0.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 0  bytes 0 (0.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 1603  bytes 148331 (148.3 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 1603  bytes 148331 (148.3 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

wlp3s0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.109.253.32  netmask 255.255.255.0  broadcast 10.109.253.255
        inet6 2402:3a80:1393:2d5b:f680:56d7:b84f:2de0  prefixlen 64  scopeid 0x0<global>
        inet6 2402:3a80:1393:2d5b:d52a:8c58:f2d4:bfb6  prefixlen 64  scopeid 0x0<global>
        inet6 fe80::e2fe:bfd9:71c2:ed2b  prefixlen 64  scopeid 0x20<link>
        ether b4:8c:9d:90:7c:37  txqueuelen 1000  (Ethernet)
        RX packets 70261  bytes 82604043 (82.6 MB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 27505  bytes 9689431 (9.6 MB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

wlx98ba5feda586: flags=4099<UP,BROADCAST,MULTICAST>  mtu 1500
        ether 98:ba:5f:ed:a5:86  txqueuelen 1000  (Ethernet)
        RX packets 0  bytes 0 (0.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 0  bytes 0 (0.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

shahid@SHAHO:~$
```

ipconfig /all for windows

Output:

```
C:\Users\SHAHO>ipconfig /all

Windows IP Configuration

   Host Name . . . . . . . . . . . . : DESKTOP-H31U0VG
   Primary Dns Suffix  . . . . . . . :
   Node Type . . . . . . . . . . . . : Hybrid
   IP Routing Enabled. . . . . . . . : No
   WINS Proxy Enabled. . . . . . . . : No

Ethernet adapter Ethernet:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Realtek PCIe GbE Family Controller
   Physical Address. . . . . . . . . : 58-11-22-3F-41-AD
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 9:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
   Physical Address. . . . . . . . . : 9A-BA-5F-ED-A5-86
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 10:

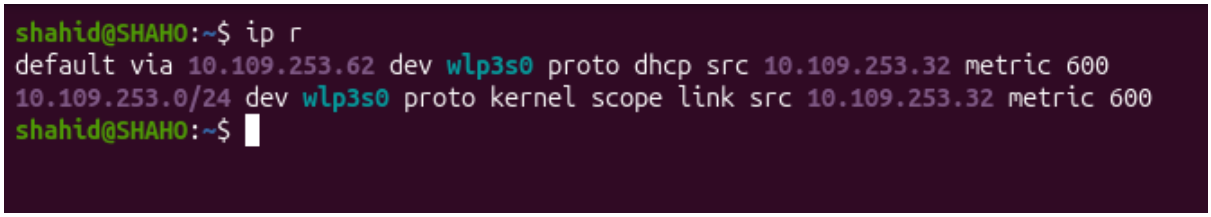
   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #2
   Physical Address. . . . . . . . . : 98-BA-5F-ED-A5-86
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
```

```
Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Realtek 8821CU Wireless LAN 802.11ac USB NIC
   Physical Address. . . . . . . . . : 98-BA-5F-ED-A5-86
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   IPv6 Address. . . . . . . . . . . : 2402:3a80:1393:2d5b:ecdb:5bc7:7e58:839c(Preferred)
   Temporary IPv6 Address. . . . . . : 2402:3a80:1393:2d5b:e8e7:816c:d5de:cd1d(Preferred)
   Link-local IPv6 Address . . . . . : fe80::4c1a:87fb:4fa:2545%10(Preferred)
   IPv4 Address. . . . . . . . . . . : 10.109.253.201(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Lease Obtained. . . . . . . . . . : 02 September 2026 2.10.11 PM
   Lease Expires . . . . . . . . . . : 02 September 2026 3.15.19 PM
   Default Gateway . . . . . . . . . : fe80::c888:96ff:fe11:5d09%10
                                       10.109.253.62
   DHCP Server . . . . . . . . . . . : 10.109.253.62
   DHCPv6 IAID . . . . . . . . . . . : 429439583
   DHCPv6 Client DUID. . . . . . . . : 00-01-01-00-31-A4-AE-29-58-11-22-3F-41-AD
   DNS Servers . . . . . . . . . . . : 10.109.253.62
                                       2402:3a80:1393:2d5b::a7
   NetBIOS over Tcpip. . . . . . . . : Enabled
```

ip r for Linux:

Output:



Ping for Windows:

Output:

```
C:\Users\SHAHO>ping google.com

Pinging google.com [2404:6800:4000:1017::71] with 32 bytes of data:
Reply from 2404:6800:4000:1017::71: time=124ms
Reply from 2404:6800:4000:1017::71: time=42ms
Reply from 2404:6800:4000:1017::71: time=40ms
Reply from 2404:6800:4000:1017::71: time=41ms

Ping statistics for 2404:6800:4000:1017::71:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 40ms, Maximum = 124ms, Average = 61ms
```

Ping for Linux:

Output:

```
shahid@SHAHO:~$ ping google.com
PING google.com (2404:6800:4000:100e::65) 56 data bytes
64 bytes from pu-in-f101.1e100.net (2404:6800:4000:100e::65): icmp_seq=1 ttl=111 time=234 ms
64 bytes from pu-in-f101.1e100.net (2404:6800:4000:100e::65): icmp_seq=2 ttl=111 time=54.0 ms
64 bytes from pu-in-f101.1e100.net (2404:6800:4000:100e::65): icmp_seq=3 ttl=111 time=383 ms
64 bytes from pu-in-f101.1e100.net (2404:6800:4000:100e::65): icmp_seq=4 ttl=111 time=396 ms
64 bytes from pu-in-f101.1e100.net (2404:6800:4000:100e::65): icmp_seq=5 ttl=111 time=1087 ms
64 bytes from pu-in-f101.1e100.net (2404:6800:4000:100e::65): icmp_seq=6 ttl=111 time=1686 ms
64 bytes from pu-in-f101.1e100.net (2404:6800:4000:100e::65): icmp_seq=7 ttl=111 time=1343 ms
64 bytes from pu-in-f101.1e100.net (2404:6800:4000:100e::65): icmp_seq=8 ttl=111 time=726 ms
64 bytes from pu-in-f101.1e100.net (2404:6800:4000:100e::65): icmp_seq=9 ttl=111 time=418 ms
64 bytes from pu-in-f101.1e100.net (2404:6800:4000:100e::65): icmp_seq=10 ttl=111 time=660 ms
64 bytes from pu-in-f101.1e100.net (2404:6800:4000:100e::65): icmp_seq=11 ttl=111 time=394 ms
64 bytes from pu-in-f101.1e100.net (2404:6800:4000:100e::65): icmp_seq=12 ttl=111 time=512 ms
64 bytes from pu-in-f101.1e100.net (2404:6800:4000:100e::65): icmp_seq=13 ttl=111 time=21.8 ms
64 bytes from pu-in-f101.1e100.net (2404:6800:4000:100e::65): icmp_seq=14 ttl=111 time=318 ms
64 bytes from pu-in-f101.1e100.net (2404:6800:4000:100e::65): icmp_seq=15 ttl=111 time=279 ms
64 bytes from pu-in-f101.1e100.net (2404:6800:4000:100e::65): icmp_seq=16 ttl=111 time=241 ms
64 bytes from pu-in-f101.1e100.net (2404:6800:4000:100e::65): icmp_seq=17 ttl=111 time=391 ms
64 bytes from pu-in-f101.1e100.net (2404:6800:4000:100e::65): icmp_seq=18 ttl=111 time=1716 ms
64 bytes from pu-in-f101.1e100.net (2404:6800:4000:100e::65): icmp_seq=19 ttl=111 time=1351 ms
64 bytes from pu-in-f101.1e100.net (2404:6800:4000:100e::65): icmp_seq=20 ttl=111 time=640 ms
64 bytes from pu-in-f101.1e100.net (2404:6800:4000:100e::65): icmp_seq=21 ttl=111 time=374 ms
64 bytes from pu-in-f101.1e100.net (2404:6800:4000:100e::65): icmp_seq=22 ttl=111 time=427 ms
64 bytes from pu-in-f101.1e100.net (2404:6800:4000:100e::65): icmp_seq=23 ttl=111 time=368 ms
64 bytes from pu-in-f101.1e100.net (2404:6800:4000:100e::65): icmp_seq=24 ttl=111 time=19.5 ms
64 bytes from pu-in-f101.1e100.net (2404:6800:4000:100e::65): icmp_seq=25 ttl=111 time=211 ms
```

tracert for windows

Output:

```
C:\Users\SHAHO>tracert google.com

Tracing route to google.com [2404:6800:4000:1017::71]
over a maximum of 30 hops:

  1     3 ms     3 ms     4 ms  2402:3a80:1393:2d5b::a7
  2    24 ms    32 ms    28 ms  fd00:192:168:173::207
  3    29 ms    33 ms    29 ms  fd00:192:168:173::207
  4   157 ms   129 ms    44 ms  fd00:192:168:203::1
  5    23 ms    15 ms    27 ms  fd00:192:168:203::2
  6    20 ms    18 ms    15 ms  2400:5200:1400:87::2
  7    34 ms    16 ms    19 ms  2402:8100:4000::7ba
  8    41 ms    41 ms    92 ms  2001:4860:1:1::ea8
  9    34 ms    41 ms    44 ms  2404:6800:8281:240::1
 10    20 ms    34 ms    42 ms  2001:4860:0:1::5ed2
 11    26 ms    37 ms    40 ms  2001:4860:0:1::879e
 12    59 ms    42 ms    40 ms  2001:4860:0:1::7743
 13    28 ms    30 ms    31 ms  2001:4860:0:1::7146
 14     *        *        *     Request timed out.
 15     *        *        *     Request timed out.
 16     *        *        *     Request timed out.
 17     *        *        *     Request timed out.
 18     *        *        *     Request timed out.
 19     *        *        *     Request timed out.
 20   763 ms    39 ms    36 ms  ly-in-f113.1e100.net [2404:6800:4000:1017::71]

Trace complete.
```

---

traceroute for Linux:

Output:

```
shahid@SHAHO:~$ traceroute google.com
traceroute to google.com (2404:6800:4000:100e::8a), 30 hops max, 80 byte packets
 1  2402:3a80:1393:2d5b::a7 (2402:3a80:1393:2d5b::a7)  21.900 ms  21.854 ms  21.795 ms
 2  fd00:192:168:173::207 (fd00:192:168:173::207)  61.842 ms  65.782 ms  61.793 ms
 3  fd00:192:168:173::207 (fd00:192:168:173::207)  77.128 ms  65.711 ms fd00:192:168:174::208 (fd00:192:168:174::208)  59.459 ms
 4  fd00:192:168:203::1 (fd00:192:168:203::1)  65.663 ms fd00:192:168:204::1 (fd00:192:168:204::1)  77.035 ms  65.617 ms
 5  fd00:192:168:203::2 (fd00:192:168:203::2)  65.594 ms * *
 6  2400:5200:1400:88::2 (2400:5200:1400:88::2)  65.502 ms  34.062 ms 2400:5200:1400:87::2 (2400:5200:1400:87::2)  34.037 ms
 7  2402:8100:4000::efe (2402:8100:4000::efe)  34.021 ms * *
 8  2001:4860:1:1::ea8 (2001:4860:1:1::ea8)  25.267 ms  31.786 ms  25.025 ms
 9  2404:6800:8285:540::1 (2404:6800:8285:540::1)  31.735 ms 2402:8100:4000::c90 (2402:8100:4000::c90)  25.147 ms  26.649 ms
10  2001:4860:0:1::5ed0 (2001:4860:0:1::5ed0)  31.656 ms 2001:4860:0:1::4fde (2001:4860:0:1::4fde)  35.156 ms 2402:8100:4000::6f1 (2402:8100:4000::6f1)  30.719 ms
11  2001:4860:0:1::9de8 (2001:4860:0:1::9de8)  30.691 ms 2001:4860:0:1::15c0 (2001:4860:0:1::15c0)  32.022 ms 2001:4860:0:1::a7ec (2001:4860:0:1::a7ec)  40.930 ms
12  2001:4860:0:1::50d7 (2001:4860:0:1::50d7)  58.154 ms 2404:6800:8281:4c0::1 (2404:6800:8281:4c0::1)  53.404 ms 2404:6800:8285:80::1 (2404:6800:8285:80::1)  56.168 ms
13  2001:4860:0:1::5aa0 (2001:4860:0:1::5aa0)  58.040 ms 2001:4860:0:1::577a (2001:4860:0:1::577a)  58.014 ms 2001:4860:0:1::542a (2001:4860:0:1::542a)  56.089 ms
14  2001:4860:0:1::a7ec (2001:4860:0:1::a7ec)  53.273 ms 2001:4860:0:1::7140 (2001:4860:0:1::7140)  53.248 ms *
15  * * *
16  * 2001:4860:0:1::577e (2001:4860:0:1::577e)  47.377 ms 2001:4860:0:1::577c (2001:4860:0:1::577c)  47.320 ms
17  * * *
18  * * *
19  * * *
20  pu-in-f138.1e100.net (2404:6800:4000:100e::8a)  28.124 ms  53.889 ms  39.098 ms
shahid@SHAHO:~$
```

nslookup for Windows:

Output:

```
PS C:\Users\student> nslookup google.com
Server:  DESKTOP-A8TKO70
Address:  fe80::6eca:ec0a:289c:91bf

Non-authoritative answer:
Name:    google.com
Addresses:  2404:6800:4000:1010::8b
          2404:6800:4000:1010::66
          2404:6800:4000:1010::71
          2404:6800:4000:1010::65
          192.178.211.100
          192.178.211.138
          192.178.211.101
          192.178.211.113
          192.178.211.102
          192.178.211.139
```

---

nslookup for Windows:

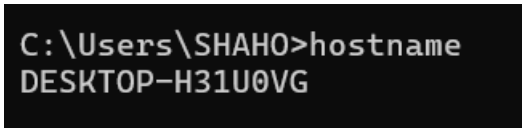
Output:

```
shahid@SHAHO:~$ nslookup google.com
Server:         127.0.0.53
Address:        127.0.0.53#53

Non-authoritative answer:
Name:   google.com
Address: 142.250.146.100
Name:   google.com
Address: 142.250.146.102
Name:   google.com
Address: 142.250.146.139
Name:   google.com
Address: 142.250.146.113
Name:   google.com
Address: 142.250.146.138
Name:   google.com
Address: 142.250.146.101
Name:   google.com
Address: 2404:6800:4000:1017::8b
Name:   google.com
Address: 2404:6800:4000:1017::66
Name:   google.com
Address: 2404:6800:4000:1017::71
Name:   google.com
Address: 2404:6800:4000:1017::65
```

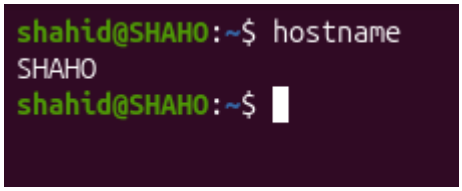
Hostname for Windows:

Output:



Hostname for Linux:

Output:



arp -a for windows:

Output:

```
C:\Users\SHAHO>arp -a

Interface: 10.109.253.201 --- 0xa
  Internet Address      Physical Address      Type
  10.109.253.62         ca-88-96-11-5d-09     dynamic
  10.109.253.255        ff-ff-ff-ff-ff-ff     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  224.77.77.77          01-00-5e-4d-4d-4d     static
  239.255.255.250       01-00-5e-7f-ff-fa     static
  255.255.255.255       ff-ff-ff-ff-ff-ff     static
```

---

arp for Linux:

Output:

```
shahid@SHAHO:~$ arp
Address                  HWtype  HWaddress           Flags Mask            Iface
_gateway                 ether   ca:88:96:11:5d:09   C                     wlp3s0
shahid@SHAHO:~$
```

CONCLUSION:

The TCP/IP utility commands such as ipconfig, ifconfig, ping, tracert, traceroute, nslookup, hostname, and arp were successfully executed and analyzed. These commands helped in understanding IP configuration, network connectivity, packet routing, DNS resolution, hostname information, and ARP details.

Questions and Answers

1. **What does the ping command do? Give an example.**

The ping command is used to check whether a device or host is reachable over a network. It sends packets to the destination and receives replies.

Example:

ping google.com

2. **What is the Linux equivalent of tracert in Windows?**

The Linux equivalent of tracert is **traceroute**.

Example:

traceroute google.com

3. **How is nslookup used in Linux?**

The nslookup command is used to find the IP address of a domain name by querying the DNS server.

Example:

nslookup google.com

4. **What does the netstat command display?**

The netstat command displays information about **network connections, open ports, and network statistics**.

Example:

Netstat

5. **What is the function of the arp command?**

The arp command is used to view and manage the **ARP table**, which contains mappings between IP addresses and MAC addresses.

Example:

arp -a

These are appropriate for the utility commands covered in your experiment and are kept short enough to write and explain in a viva.